Data Setup & Initial EDA Report

Week 1 – Data Integration & Foundational Analysis

Data Visualization Associate Internship — Excelerate
Prepared by: Sudheer Muthyala
Team Name: Team - 34
Datasets: Opportunity Data | Learner Data

1. Database Setup

1.1 Objective

The objective of Week 1 was to set up a PostgreSQL database environment, import the Opportunity and Learner datasets, connect the database to Looker Studio, perform initial exploratory analysis using SQL, and identify data quality issues before building dashboards.

1.2 PostgreSQL Environment

PostgreSQL was hosted using Supabase, which provides a managed PostgreSQL database in the cloud. Supabase was selected because it offers an integrated SQL Editor, secure cloud storage, and direct compatibility with Google Looker Studio using the PostgreSQL connector.

Connection Method:

- Database: PostgreSQL
- Platform: Supabase
- Visualization Tool: Looker Studio
- SQL Interface: Supabase SQL Editor

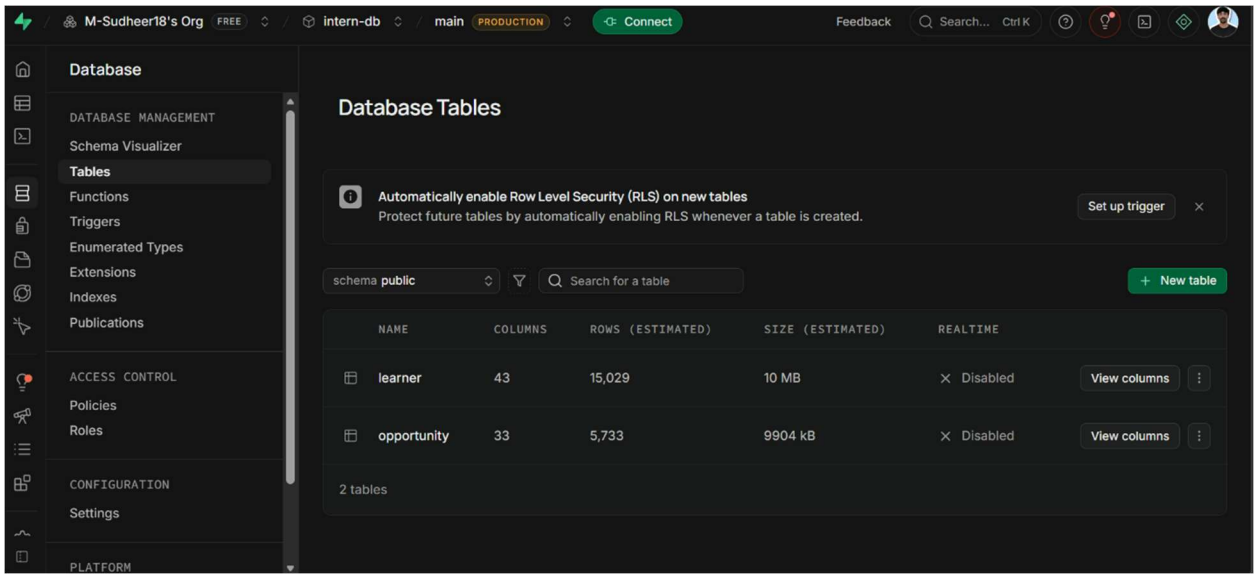


Fig 1: Supabase PostgreSQL Connection.

1.3 Dataset Upload

Two datasets were provided for analysis.

Dataset	Rows	Columns
Learner	15,029	43
Opportunity	5,733	33

Both datasets were imported into PostgreSQL using Supabase. The data was successfully loaded into two tables named **learner** and **opportunity**.

Learner and Opportunity tables successfully imported into PostgreSQL.

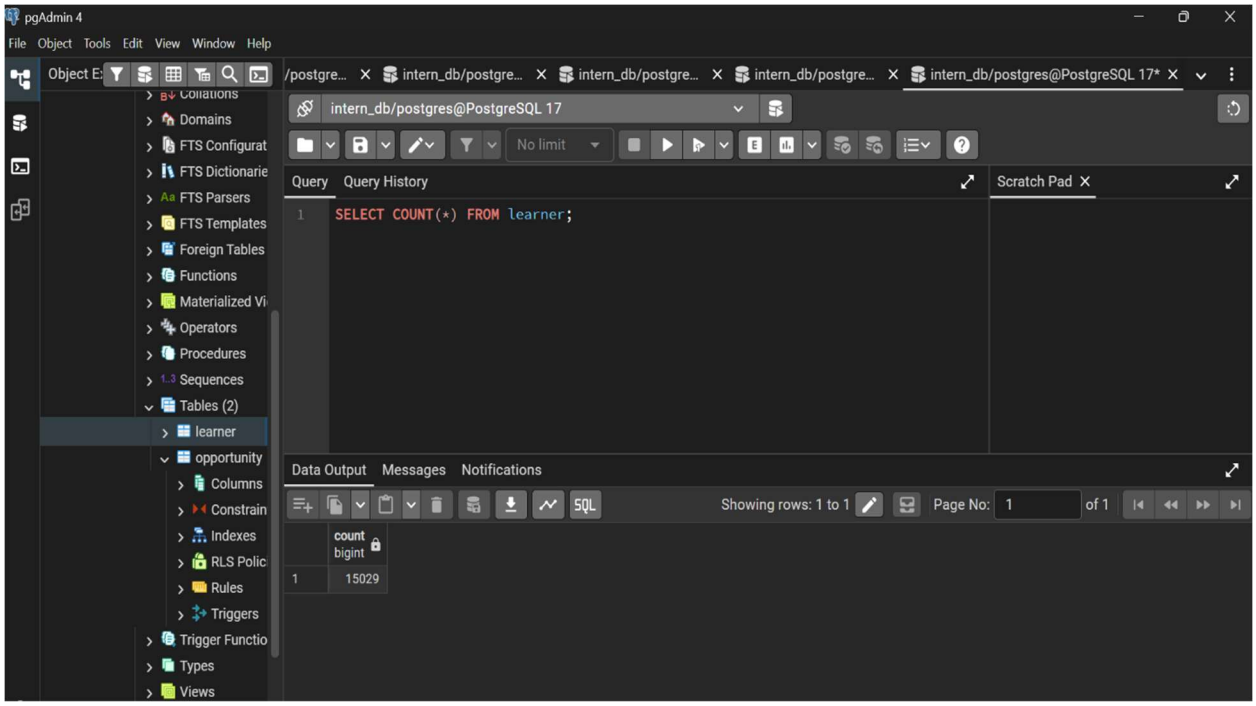


Fig 2: Learner Data Loaded

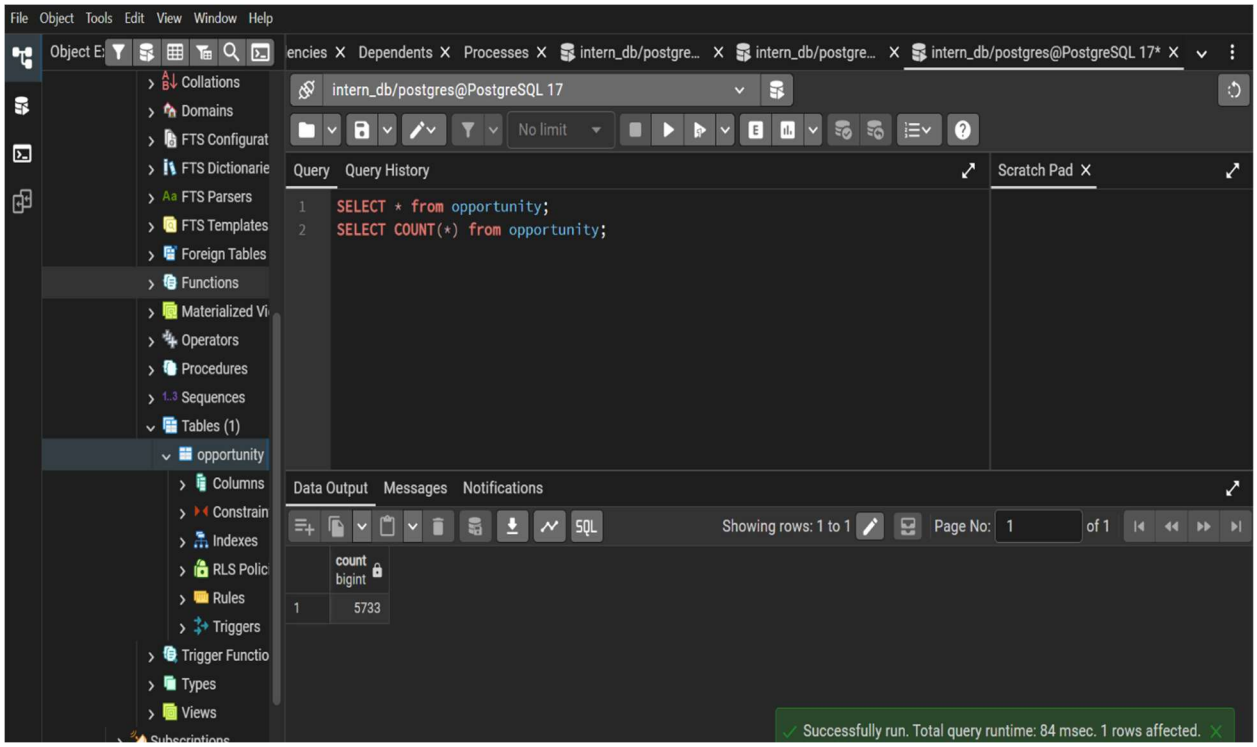


Fig 3: Opportunity Data Loaded

1.4 Data Verification

The imported data was verified by checking the total number of rows available in each table.

SELECT COUNT(*) FROM learner;

SELECT COUNT(*) FROM opportunity;

Table	**Rows**
learner	15,029
opportunity	5,733

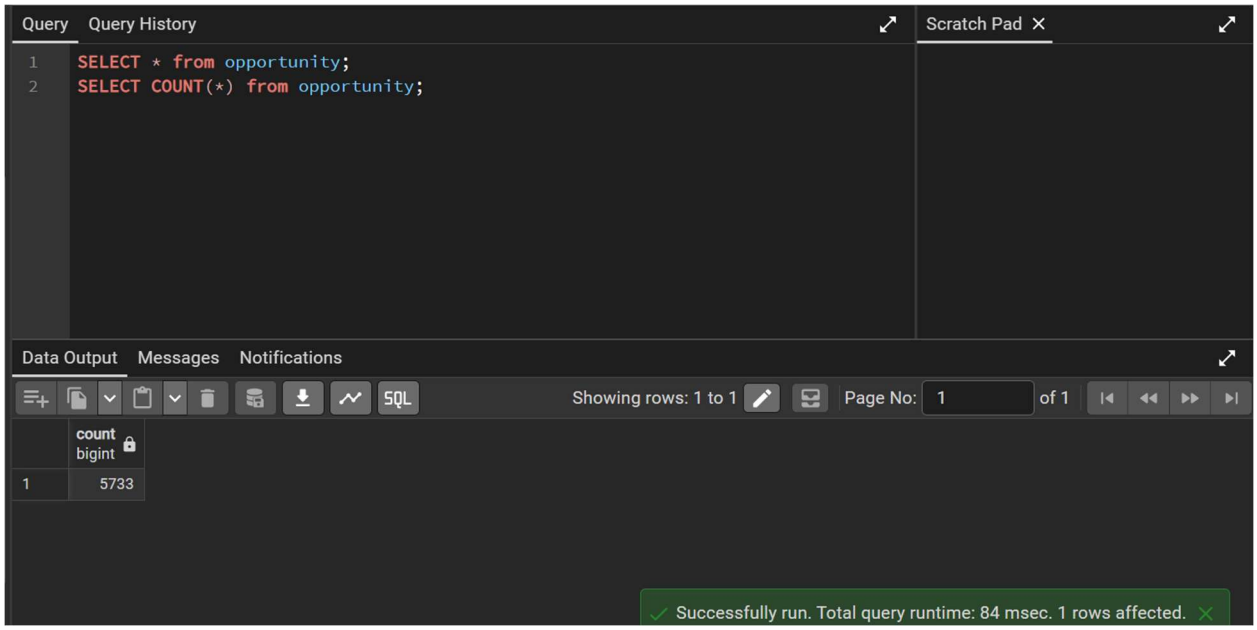


Fig 4: SQL verification of imported records.

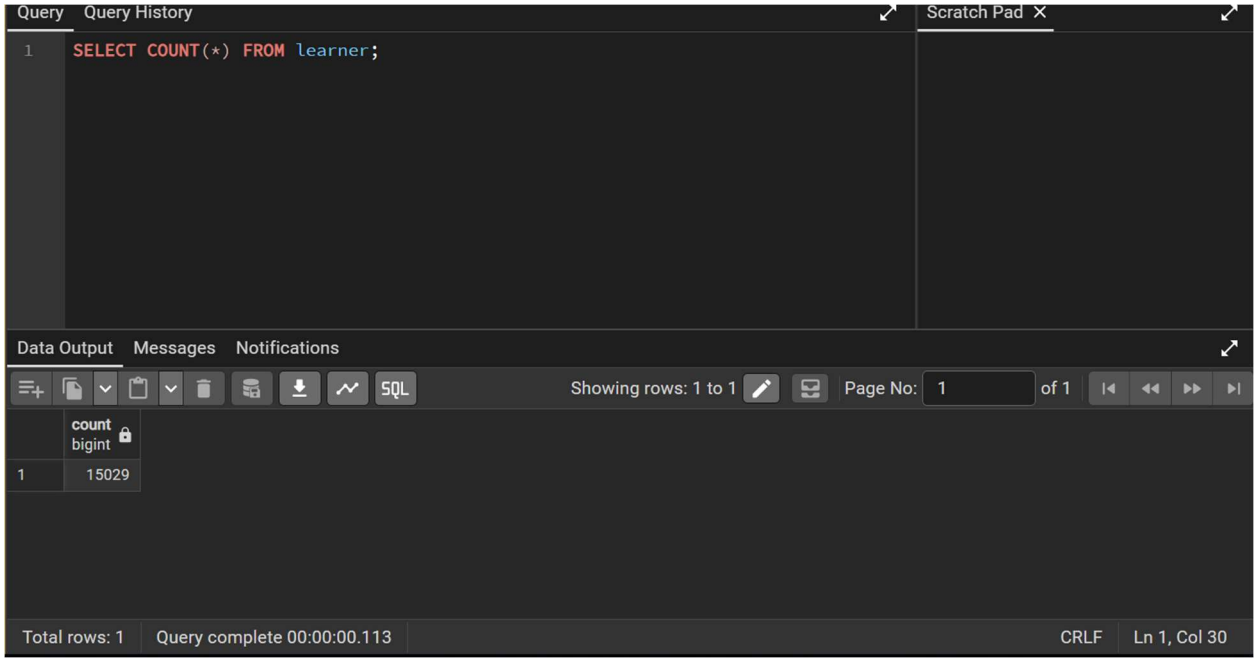


Fig 5: SQL verification of imported records

1.5 Table Creation in PostgreSQL

Before migrating the datasets to Supabase, the required database tables were created in the local PostgreSQL environment. Separate tables were created for the **learner** and **opportunity** datasets based on the provided CSV files. After the table structures were created, the datasets were imported into PostgreSQL and verified. The data was then migrated to Supabase PostgreSQL for cloud-based access and integration with Looker Studio.

The following tables were created:

Table Name	Purpose
learner	Stores learner application and enrollment information.
opportunity	Stores opportunity details such as internships, courses, competitions, events, and job simulations.

The table schemas were created using the **PostgreSQL Query Tool in pgAdmin 4**. After the tables were created, the corresponding CSV datasets were imported into each table. Finally, the PostgreSQL database was migrated to Supabase for cloud-based access and reporting.

Example SQL used to create a table:

```
CREATE TABLE learner (
    pk TEXT,
    application_id TEXT,
    assigned_cohort TEXT,
    category TEXT,
    status TEXT,
    apply_date TEXT,
    completion_date TEXT,
    ...
);
CREATE TABLE opportunity (
    opportunity_id TEXT,
    category TEXT,
    badge TEXT,
    duration TEXT,
    fee TEXT,
    location TEXT,
    ... );
```

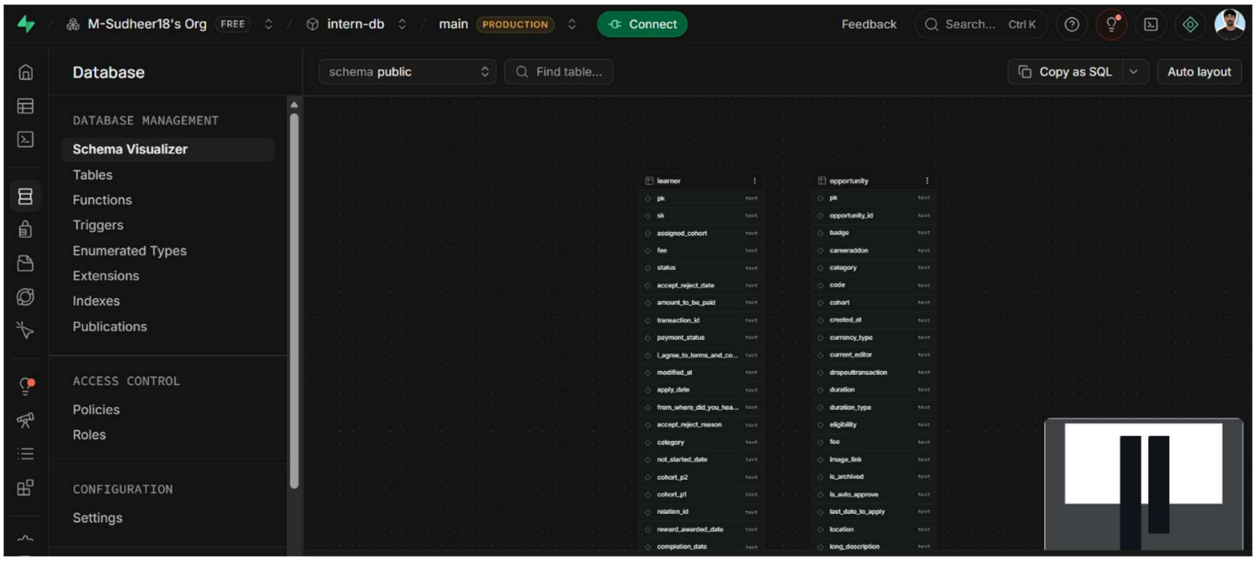


Fig 6: Schema visualization of the Learner and Opportunity tables in Supabase PostgreSQL.

The schema visualization confirms that both the **Learner** and **Opportunity** tables were successfully migrated to Supabase PostgreSQL and are available for querying and visualization through Looker Studio.

1.6 Connection to Looker Studio

After the datasets were successfully imported into Supabase PostgreSQL, the database was connected to Google Looker Studio using the PostgreSQL connector. The connection was configured using the Supabase host, database name, username, password, and SSL settings. Once the connection was established, both the **learner** and **opportunity** tables became available as data sources for creating dashboards and visualizations. This integration enables real-time reporting and interactive analysis directly from the PostgreSQL database.

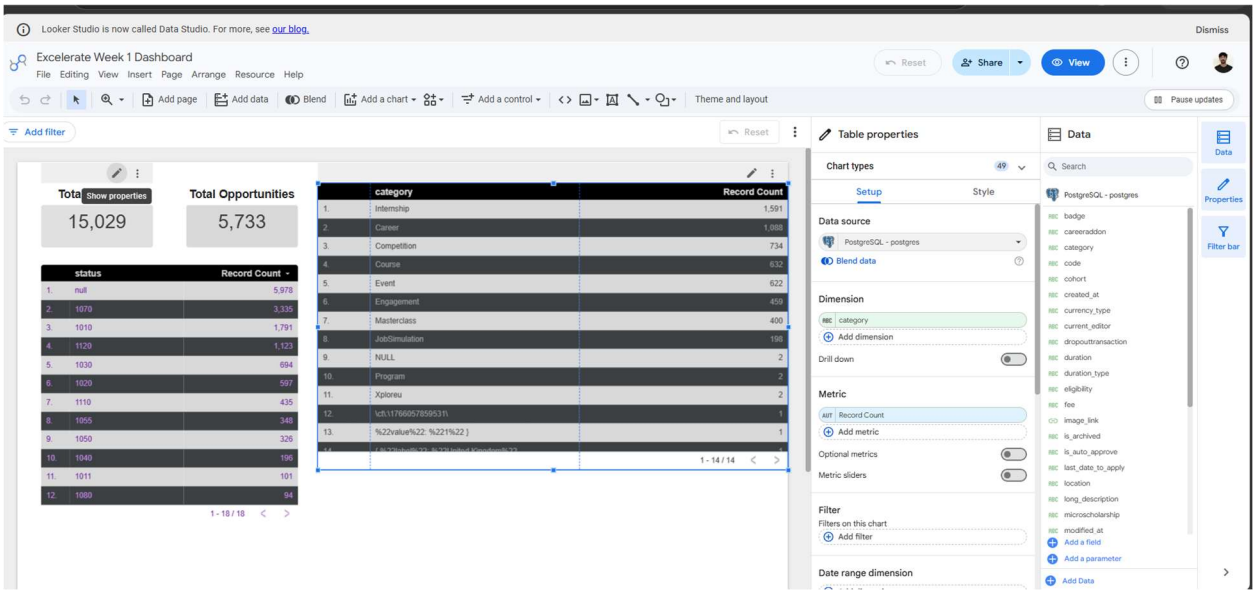


Fig 7: Supabase PostgreSQL successfully connected to Google Looker Studio.

2. Dataset Overview

2.1 Introduction

The internship provided two datasets, **Learner** and **Opportunity**, which serve as the foundation for the analysis. Before performing exploratory analysis and building dashboards, both datasets were examined to understand their structure, identify key variables, and verify the imported records. This step helped in understanding the relationships between the datasets and identifying fields that would be useful for future analysis.

2.2 Dataset Summary

Dataset	Rows	Columns	Description
Learner	15,029	43	Contains learner registration, application, cohort, completion, and engagement details.
Opportunity	5,733	33	Contains opportunity information such as internships, courses, competitions, events, and job simulations.

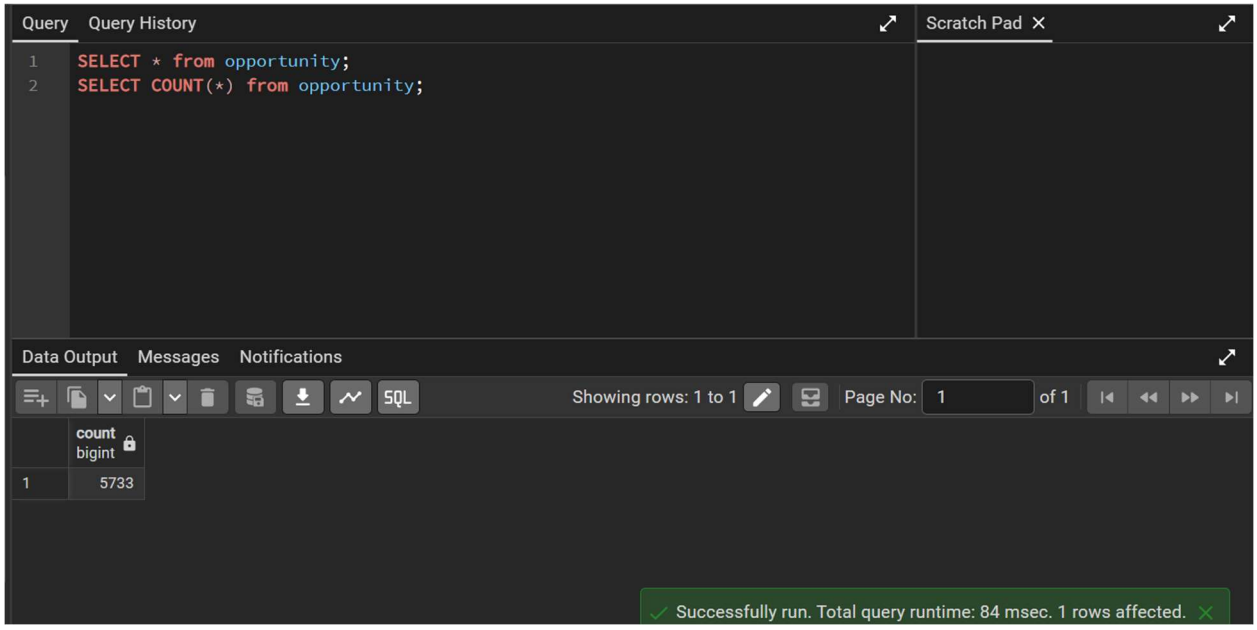


Fig 8: Summary of the Opportunity datasets in Supabase PostgreSQL

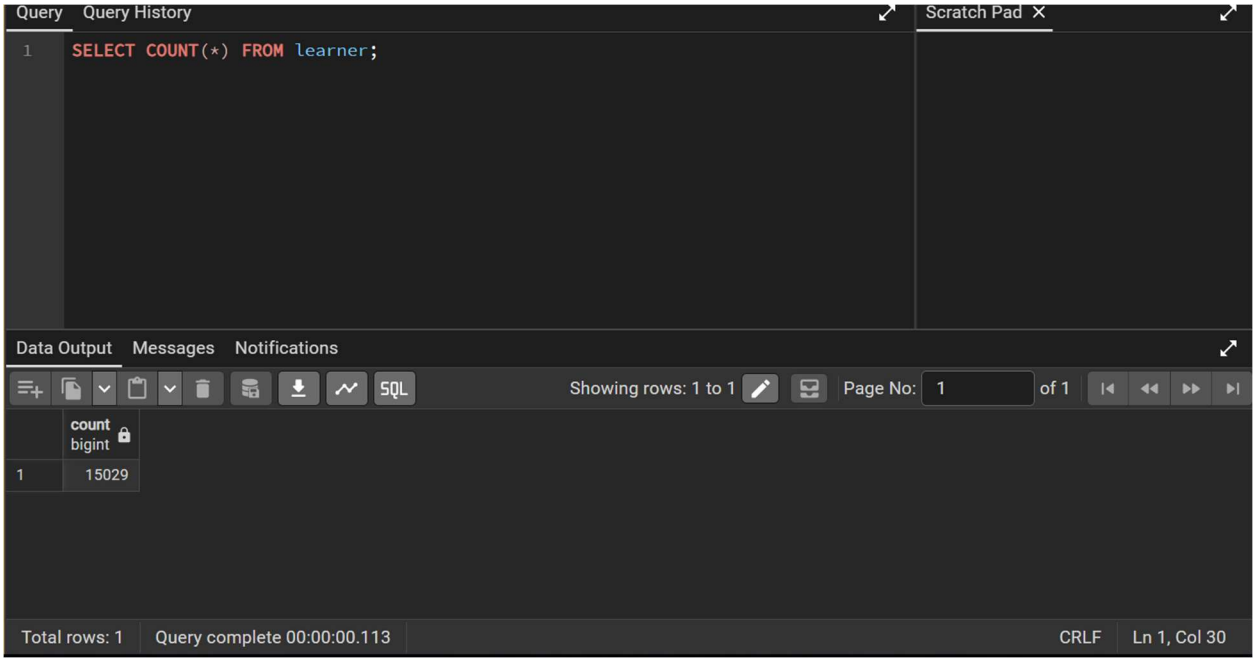


Fig 9: Summary of the Learner datasets in Supabase PostgreSQL

2.3 Key Tables and Important Fields

Learner Table

Field	**Description**
pk	Unique learner identifier
application_id	Application ID
assigned_cohort	Assigned learner cohort
category	Learning category
status	Current learner status
apply_date	Application date
completion_date	Completion date
created_at	Record creation timestamp

Opportunity Table:

Field	**Description**
opportunity_id	Unique opportunity identifier
category	Opportunity category
badge	Badge associated with opportunity
duration	Opportunity duration
duration_type	Duration unit
fee	Opportunity fee
location	Opportunity location
eligibility	Eligibility requirements
created_at	Record creation timestamp

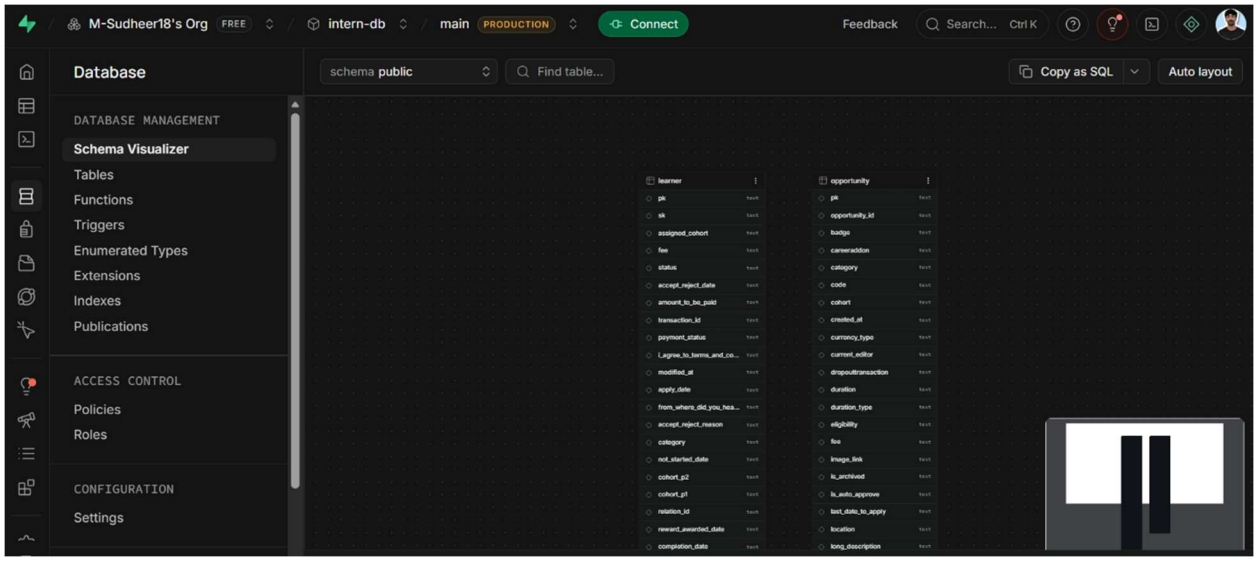


Fig 10: Schema visualization showing the key fields in the Learner and Opportunity tables.

2.4 Initial Observations

The Learner dataset contains considerably more records than the Opportunity dataset, indicating that multiple learners can participate in a single opportunity. The Learner table focuses on learner-related information such as applications, cohorts, completion, and engagement, whereas the Opportunity table stores descriptive information about available opportunities. Several categorical variables, including **status**, **category**, **cohort**, **location**, and **duration type**, are expected to play an important role in subsequent exploratory analysis and dashboard development.

2.5 Important Variables for Future Analysis

Variable	**Dataset**
status	Learner
category	Both
assigned_cohort	Learner
completion_date	Learner
apply_date	Learner
duration_type	Opportunity
location	Opportunity
fee	Opportunity

Section 3. Exploratory Analysis

3.1 Dataset Size Verification

Verify the number of records in each dataset.

```
SELECT
  (SELECT COUNT(*) FROM learner) AS total_learners,
  (SELECT COUNT(*) FROM opportunity) AS total_opportunities;
```

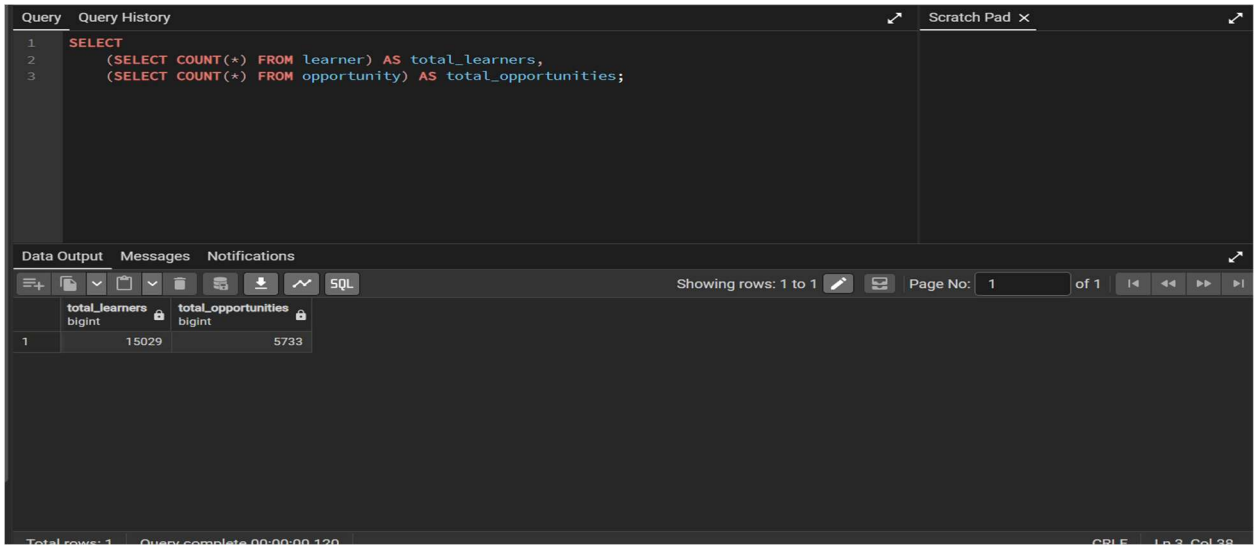
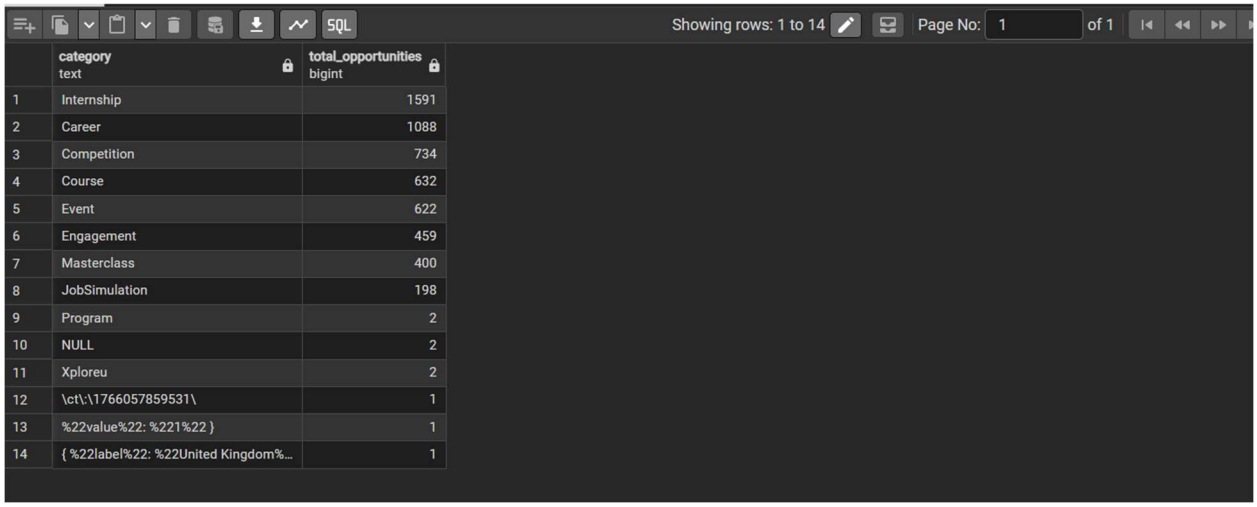


Fig 11: Total records available in the Learner and Opportunity datasets.

3.2: Distribution of Opportunities by Category

This analysis examines how opportunities are distributed across different categories. Understanding the category distribution provides insight into the types of opportunities available in the dataset and highlights areas with greater participation.

```
SELECT
    category,
    COUNT(*) AS total_opportunities
FROM opportunity
GROUP BY category
ORDER BY total_opportunities DESC;
```



	category text	total_opportunities bigint
1	Internship	1591
2	Career	1088
3	Competition	734
4	Course	632
5	Event	622
6	Engagement	459
7	Masterclass	400
8	JobSimulation	198
9	Program	2
10	NULL	2
11	Xploreu	2
12	\ct\:\1766057859531\	1
13	%22value%22: %221%22 }	1
14	{ %22label%22: %22United Kingdom%...	1

Fig 12: Distribution of opportunities across different categories.

3.3 Opportunity Currency Distribution

```
SELECT
    currency_type,
    COUNT(*) AS total_opportunities
FROM opportunity
GROUP BY currency_type
ORDER BY total_opportunities DESC;
```

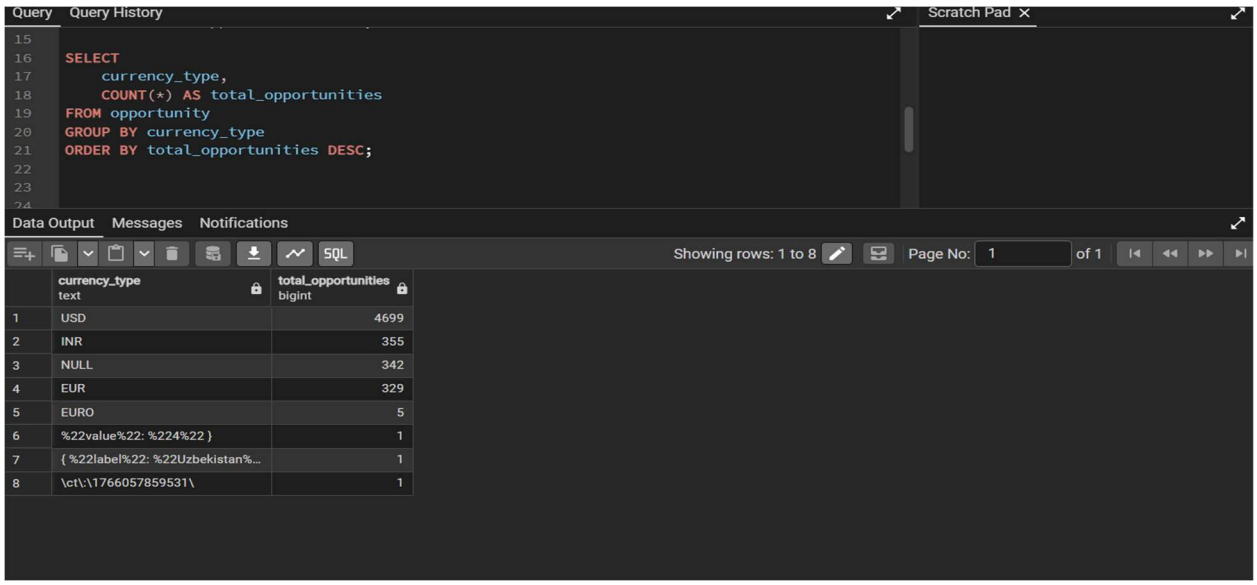


Fig 13: Distribution of opportunities by currency type.

3.4 Learner Status Distribution

```
SELECT
    status,
    COUNT(*) AS total_learners
FROM learner
GROUP BY status
ORDER BY total_learners DESC;
```

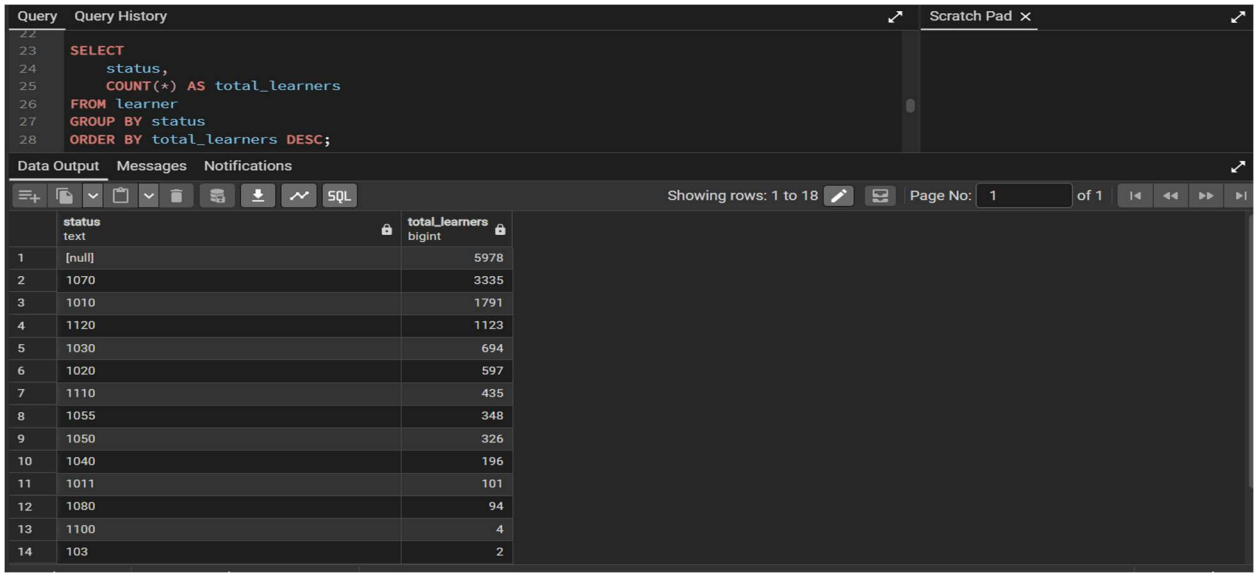


Fig 14: Distribution of learners by status.

3.5: Learner Payment Status Distribution

This analysis examines the distribution of learner payment statuses to understand the overall payment completion trends and identify the most common payment states.

```
SELECT
    payment_status,
    COUNT(*) AS total_learners
FROM learner
GROUP BY payment_status
ORDER BY total_learners DESC;
```

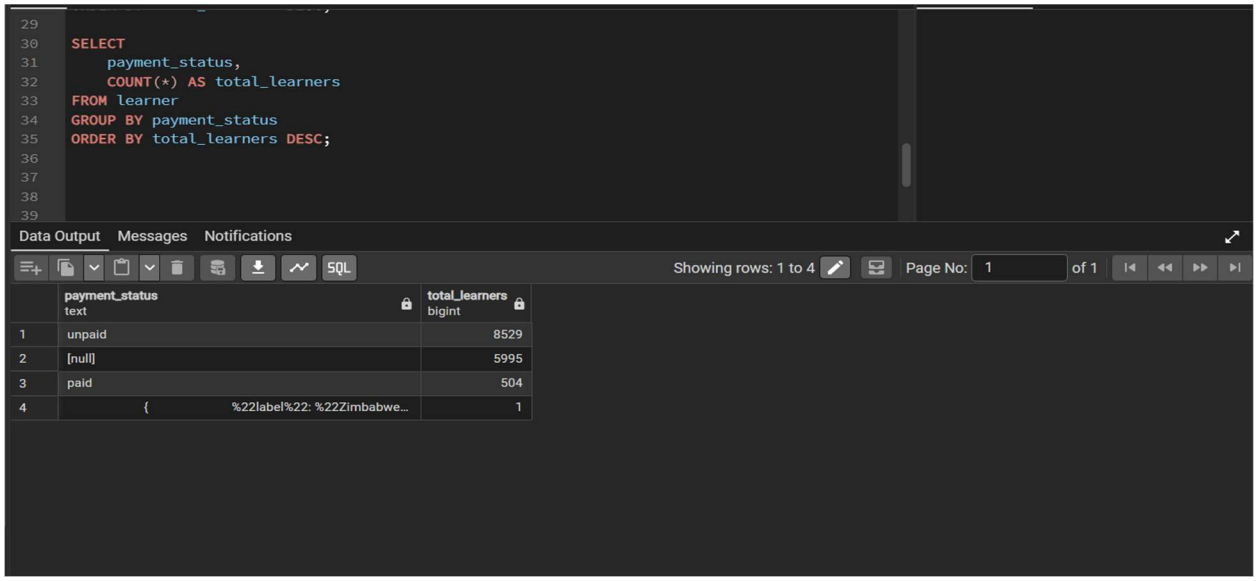


Fig 15: Distribution of learners based on payment status.

4. Data Quality Assessment

4.1 Missing Values

The first step in assessing data quality is identifying missing values in important columns. Missing data can reduce the completeness of the dataset and may affect the reliability of statistical analysis and dashboard visualizations.

```
SELECT
    COUNT(*) FILTER (WHERE category IS NULL) AS missing_category,
    COUNT(*) FILTER (WHERE currency_type IS NULL) AS missing_currency,
    COUNT(*) FILTER (WHERE role IS NULL) AS missing_role
FROM opportunity;
```

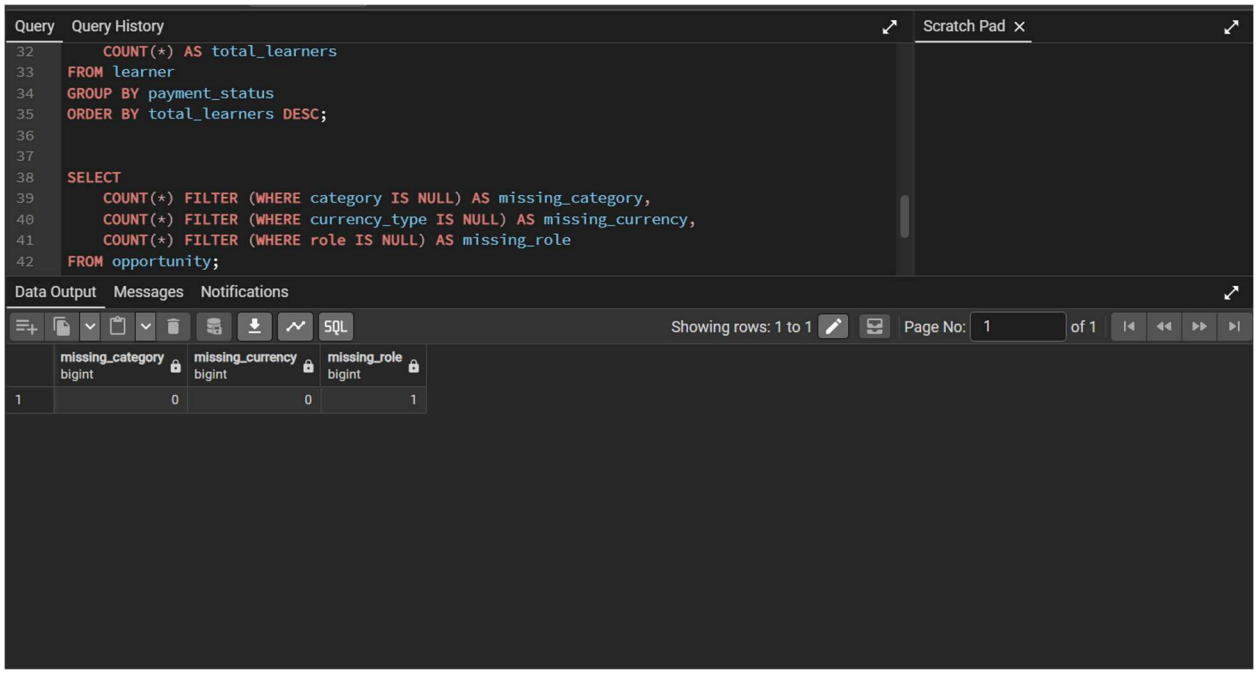


Fig 16: Missing values identified in the Opportunity dataset.

```
SELECT
    COUNT(*) FILTER (WHERE status IS NULL) AS missing_status,
    COUNT(*) FILTER (WHERE payment_status IS NULL) AS missing_payment_status,
    COUNT(*) FILTER (WHERE assigned_cohort IS NULL) AS missing_assigned_cohort
FROM learner;
```

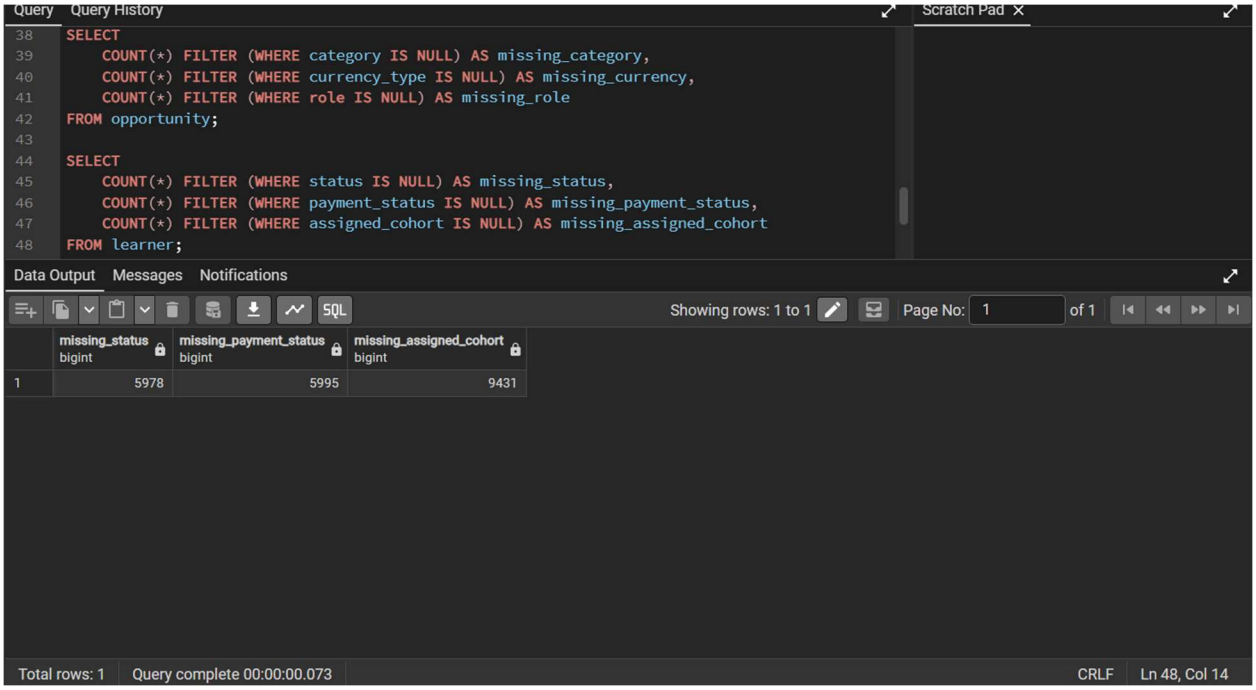


Fig 17: Missing values identified in the Learner dataset.

The datasets were examined for missing values in selected key attributes. The analysis identified null values in important fields such as category, currency type, and role in the Opportunity dataset, as well as status, payment status, and assigned cohort in the Learner dataset. These missing values may influence aggregation, filtering, and visualization results if they are not addressed during the data preparation stage. Appropriate data cleaning or imputation techniques should be considered before performing advanced analysis.

4.2 Duplicate Record Assessment

Duplicate records can introduce bias into the analysis by causing the same information to be counted multiple times. Therefore, both datasets were checked for duplicate primary identifiers.

```
SELECT
   opportunity_id,
   COUNT(*) AS duplicate_count
FROM opportunity
GROUP BY opportunity_id
HAVING COUNT(*) > 1;
```

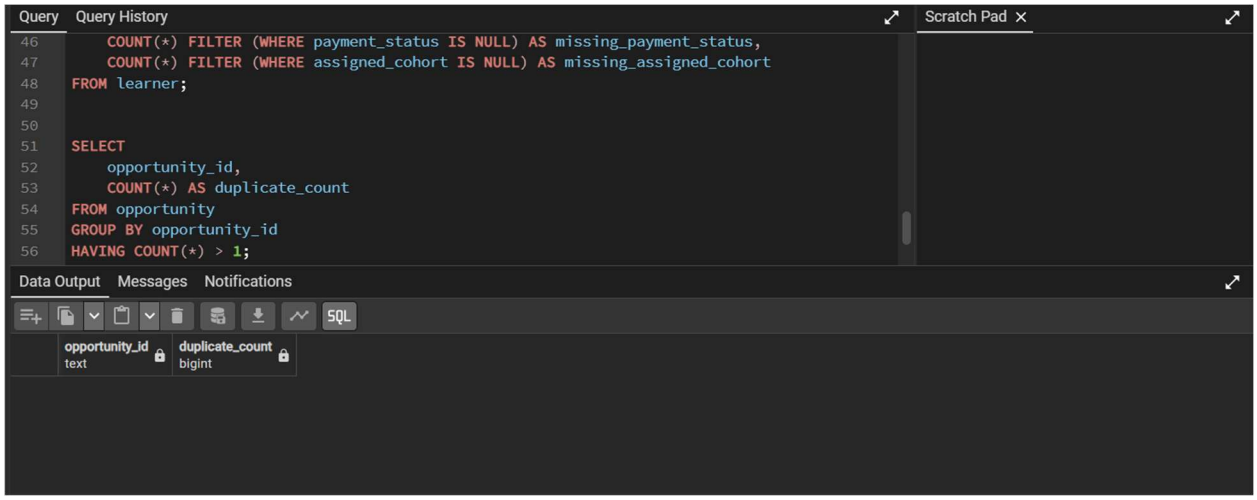


Fig 18: Duplicate record check for the Opportunity dataset.

```
SELECT
    pk,
    COUNT(*) AS duplicate_count
FROM learner
GROUP BY pk
HAVING COUNT(*) > 1;
```

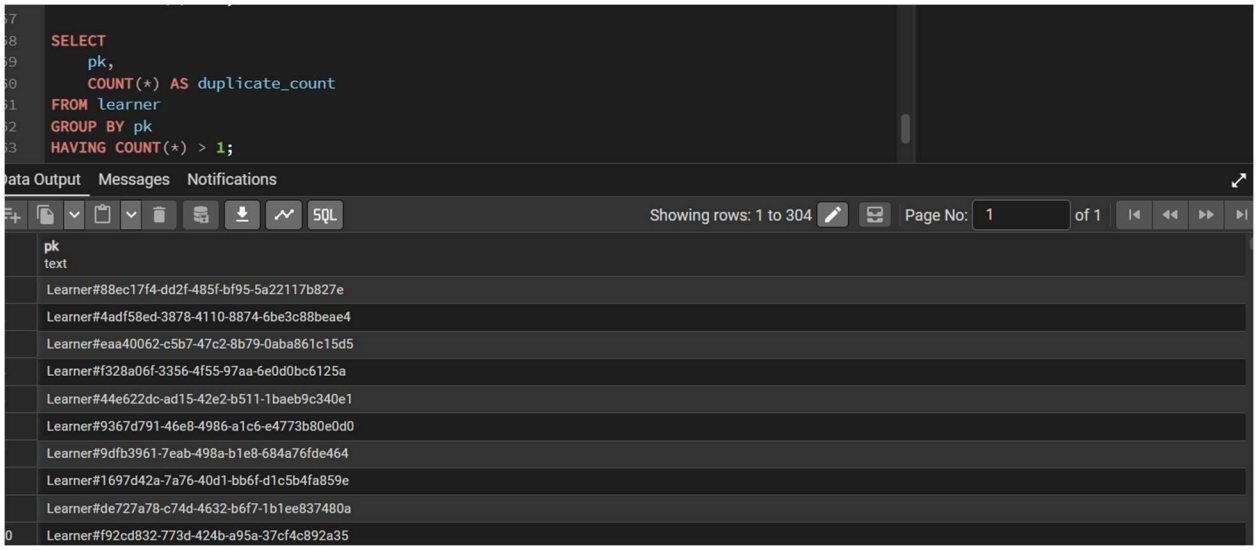


Fig 19: Duplicate record check for the Learner dataset.

Duplicate record analysis was performed using the primary identifiers of both datasets. The Opportunity dataset showed no duplicate opportunity_id values, indicating that each opportunity record is unique. However, the Learner dataset returned duplicate pk values, suggesting that some learner records share the same identifier. These duplicate records may lead to inaccurate counts, repeated learner entries, and biased analytical results if they are not removed or investigated before further analysis.

4.3 Inconsistent Formatting

```
SELECT DISTINCT category
FROM opportunity
ORDER BY category;
```

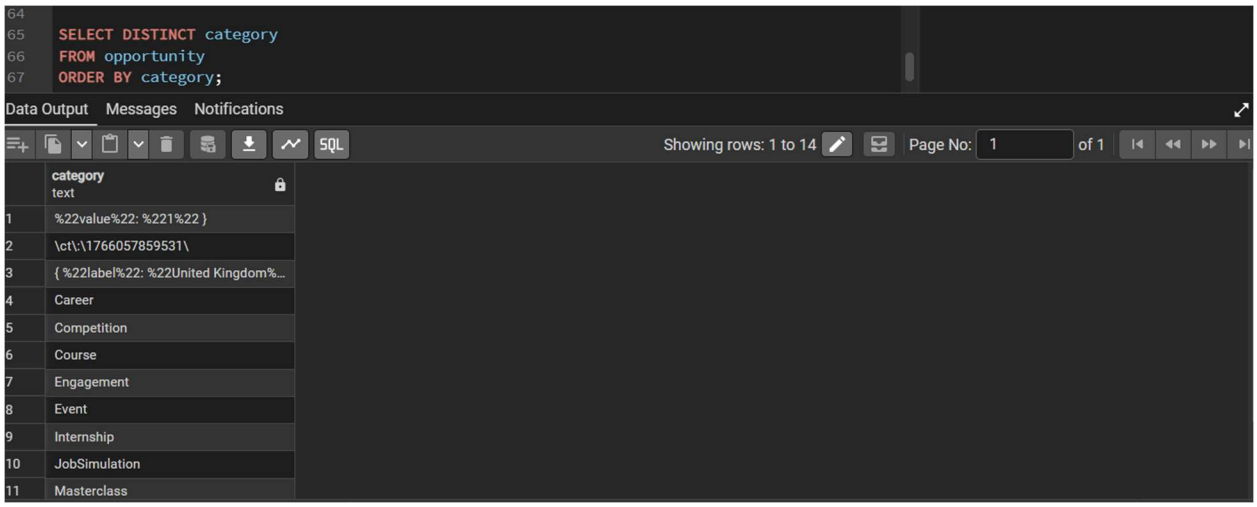


Fig 20: Distinct category values in the Opportunity dataset highlighting inconsistent formatting.

The categorical values in the Opportunity dataset were reviewed to identify inconsistencies. The results revealed variations in category values, including null entries and unexpected text values. Such inconsistencies may affect grouping, filtering, and dashboard visualizations. Standardizing categorical values during data cleaning would improve data quality and ensure consistent analytical results.

4.4 Unusual or Unexpected Values

The datasets were examined to identify unusual or unexpected values that could affect the quality and accuracy of future analyses. This helps detect invalid entries, inconsistent categories, and records requiring further cleaning.

```
SELECT
    category,
    COUNT(*) AS total_records
FROM opportunity
GROUP BY category
ORDER BY total_records DESC;
```

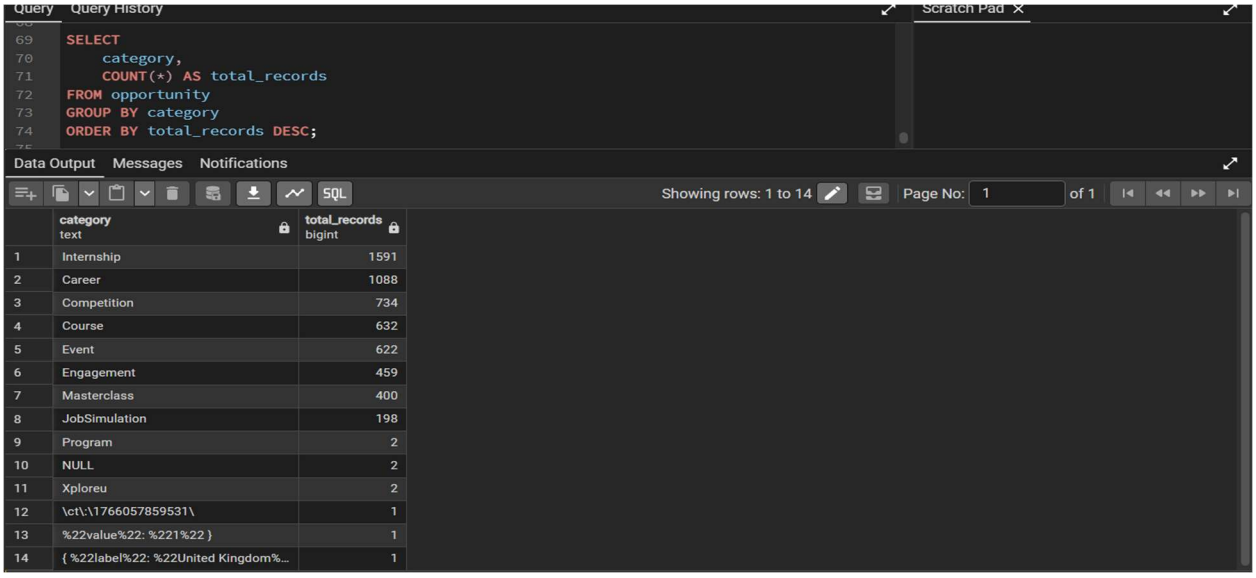


Fig 21: Unusual and unexpected values identified in the Opportunity dataset.

```
SELECT
   payment_status,
   COUNT(*) AS total_records
FROM learner
GROUP BY payment_status
ORDER BY total_records DESC;
```

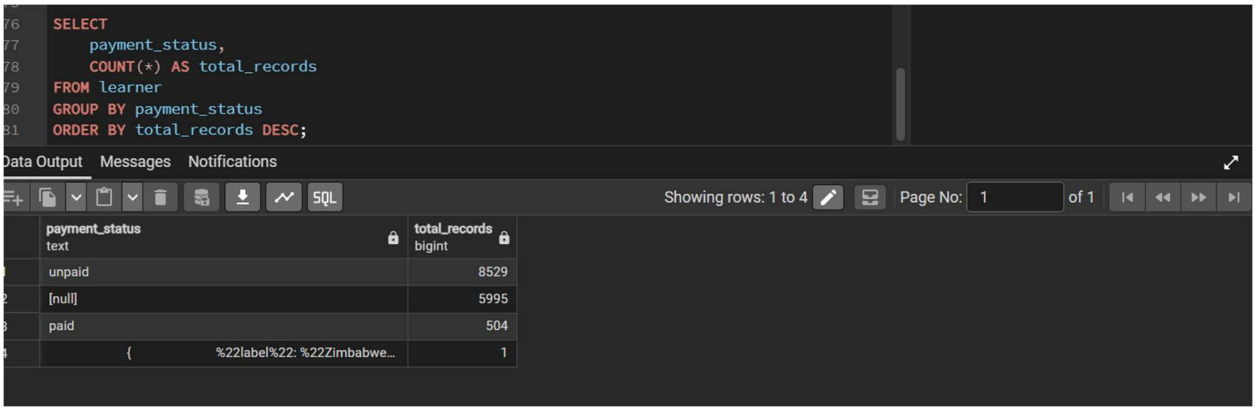


Fig 22: Unusual and unexpected values identified in the Learner dataset

The exploratory analysis revealed several unusual and unexpected values in both datasets. The Opportunity dataset contained null entries and a small number of inconsistent category values that differed from the standard categories. Similarly, the Learner dataset contained unexpected payment status values and missing entries. These anomalies may affect grouping, filtering, and summary statistics if left unaddressed. Standardizing categorical values and handling missing or invalid records during the data preparation phase will improve the reliability of future analyses and dashboard visualizations.

5. Conclusion

During Week 1 of the Excelerate Data Visualization Internship, a PostgreSQL database environment was successfully established, and the Learner and Opportunity datasets were imported into the database. The datasets were migrated to Supabase PostgreSQL and connected to Looker Studio for future dashboard development.

Initial exploratory data analysis was performed using SQL queries to examine dataset size, record distributions, learner status, payment status, opportunity categories, and currency types. The analysis also identified several data quality issues, including missing values, duplicate learner records, inconsistent formatting, and unexpected categorical values. These findings provide a strong foundation for data cleaning, transformation, and dashboard development in the subsequent weeks of the internship.